

Zachary Nolan Sunberg

Ann and H. J. Smead Aerospace Engineering Sciences
University of Colorado Boulder
720-933-7799 • zachary.sunberg@colorado.edu • zachary.sunberg.net

Academic Appointments

Assistant Professor [January 2020 – Present]
University of Colorado, Boulder, CO
Ann and H. J. Smead Aerospace Engineering Sciences Department
Computer Science Department, by courtesy

Postdoctoral Research Scholar [October 2018 – October 2019]
University of California, Berkeley, CA
Supervisor: Claire Tomlin, Hybrid Systems Laboratory

Education

Doctor of Philosophy in Aeronautics and Astronautics [2018]
Stanford University, Stanford, CA
Advisor: Mykel Kochenderfer | Thesis: "Safety and Efficiency in Autonomous Vehicles through Planning with Uncertainty"

Master of Science in Aerospace Engineering [2013]
Texas A&M University, College Station, TX
Advisor: Jonathan Rogers | Thesis: "A Real Time Expert Control System for Helicopter Autorotation"

Bachelor of Science in Aerospace Engineering [2011]
Texas A&M University, College Station, TX
Summa cum Laude, Minor in Mathematics

Industry Experience

Google, Inc., Mountain View, CA [Summer 2014]
Software Engineering Intern
Evaluated and optimized a NASA collision avoidance program for use with Google self-piloted air vehicles.

Lockheed Martin Autonomous Systems, Littleton, CO [Summer 2009]
Engineering Intern
Helped in testing of autonomous SMSS allterrain military transport vehicle navigation system; wrote rough terrain navigation program in C++ based on the A* search algorithm; wrote software in C++ for analyzing the performance of an advanced video analysis tool.

Awards and Fellowships

Smead Aerospace Engineering Department Outstanding Junior Faculty Award [2025]
NSF CAREER Award [2024]
IET Cyberphysical Systems Journal Premium Paper Award [2023]
AIAA Journal of Guidance, Control, and Dynamics Excellent Reviewer [2021,2022]
IJCAI 2019 Distinguished Program Committee member [2019]
Association for the Advancement of Artificial Intelligence (AAAI) Doctoral Consortium [2018]
National Science Foundation Graduate Research Fellowship [2012-2016]

Publications

My name and the names of my students are indicated with bold font.

Peer Reviewed Journal Articles.....

- [J1] **[In Revision]** Benjamin D. Kraske, Qi Heng Ho, Federico Rossi, Morteza Lahijanian, and **Zachary Sunberg**. “Robustness of POMDP Policies to Observation Perturbations”. In: *Journal of Artificial Intelligence Research* (2026). In preparation for resubmission
- [J2] **[Under Review]** Ofer Dagan, Tyler Becker, and **Zachary N. Sunberg**. “Hypothesis-Driven Sensor Tasking for Space Domain Awareness”. In: *The Journal of the Astronautical Sciences* (2026). Under review
- [J3] **[In Revision]** Mark Boyer, Adrian Robinson, Torin K. Clark, and **Zachary Sunberg**. “Comprehensive Evaluation of Explanation Types in a Spaceflight-Relevant Human-Autonomy Teaming Task”. In: *IEEE Transactions on Human Factors and Ergonomics* (2026). In preparation for resubmission
- [J4] **[Under Review]** Austin Monell and **Zachary Sunberg**. “Trajectory Optimization and Performance of a V-Twin Airborne Wind Energy System”. In: *Wind Energy Science* (2026). Under Review
- [J5] Prashin Sharma, Benjamin Kraske, Joseph Kim, **Zakariya Laouar**, Ella Atkins, and **Zachary Sunberg**. “Risk-Aware Markov Decision Process Contingency Management Autonomy for Uncrewed Aircraft Systems”. In: *AIAA Journal of Aerospace Information Systems (JAIS)* 21.3 (2024). doi: 10.2514/1.I011235. URL: [🔗](#)
- [J6] Qi Heng Ho, **Zachary N. Sunberg**, and Morteza Lahijanian. “Sampling-based Reactive Synthesis for Nondeterministic Hybrid Systems”. In: *Robotics and Automation Letters (RA-L)* 9.2 (2024). doi: 10.1109/LRA.2023.3340029. URL: [🔗](#)
- [J7] Michael H. Lim, Tyler J. Becker, Mykel J. Kochenderfer, Claire J. Tomlin, and **Zachary N. Sunberg**. “Optimality Guarantees for Particle Belief Approximation of POMDPs”. In: *Journal of Artificial Intelligence Research* 77 (2023). doi: 10.1613/jair.1.14525. URL: [🔗](#)
- [J8] Benjamin W. Blonder, Michael H. Lim, **Zachary Sunberg**, and Claire Tomlin. “Navigation between initial and desired community states using shortcuts”. In: *Ecology Letters* 26.4 (2023). doi: 10.1111/ele.14171. URL: [🔗](#)
- [J9] **Zachary Sunberg** and Mykel Kochenderfer. “Improving Automated Driving through Planning with Human Internal States”. In: *IEEE Transactions on Intelligent Transportation Systems* 23.11 (2022). doi: 10.1109/TITS.2022.3182687. URL: [🔗](#)
- [J10] Shakeeb Ahmad, **Zachary N. Sunberg**, and J. Sean Humbert. “End-to-End Probabilistic Depth Perception and 3D Obstacle Avoidance using POMDP”. in: *Journal of Intelligent & Robotic Systems* 103.2 (2021), pp. 1–18. doi: 10.1007/s10846-021-01489-w
- [J11] Patrick Slade, **Zachary Sunberg**, and Mykel J. Kochenderfer. “Estimation and Control Using Sampling-Based Bayesian Reinforcement Learning”. In: *IET Cyber-Physical Systems: Theory and Applications* 5 (1 2020). doi: 10.1049/iet-cps.2019.0045. URL: [🔗](#)
- [J12] Maxim Egorov, **Zachary N. Sunberg**, Edward Balaban, Tim A. Wheeler, Jayesh K. Gupta, and Mykel J. Kochenderfer. “POMDPs.jl: A Framework for Sequential Decision Making under Uncertainty”. In: *Journal of Machine Learning Research* 18.26 (2017), pp. 1–5. URL: [🔗](#)
- [J13] **Zachary Sunberg**, Suman Chakravorty, and Richard Scott Erwin. “Information Space Receding Horizon Control for Multisensor Tasking Problems”. In: *IEEE Transactions on Cybernetics* 46.6 (2016), pp. 1325–1336. URL: [🔗](#)
- [J14] **Zachary N. Sunberg**, Nathaniel R. Miller, and Jonathan D. Rogers. “A Real-Time Expert Control System For Helicopter Autorotation”. In: *Journal of the American Helicopter Society* 60.2 (2015), pp. 1–15. ISSN: 2161-6027. doi: 10.4050/JAHS.60.022008. URL: [🔗](#)
- [J15] **Zachary Sunberg**, Suman Chakravorty, and Richard Scott Erwin. “Information Space Receding Horizon Control”. In: *IEEE Transactions on Cybernetics* 43.6 (2013), pp. 2255–2260. doi: 10.1109/TSMCB.2012.2236313. URL: [🔗](#)

- [J16] **Zachary Sunberg** and Jonathan Rogers. “A Belief Function Distance Metric for Orderable Sets”. In: *Information Fusion* 14.4 (2013), pp. 361–373. ISSN: 1566-2535. DOI: 10.1016/J.INFFUS.2013.03.003. URL: 

Peer Reviewed Journal-Equivalent Conference Publications

Journal-equivalent conference publications have rigorous peer review of the entire article, an acceptance rate of approximately 30% or less, and are recognized to be as important as journal papers in the field.

- [JC1] [Under Review] **Lisa Ventura, Tyler Becker, and Zachary Sunberg**. “Constraint Budget Allocation in Multiagent MDPs”. In: *Robotics: Science and Systems (RSS)*. Under review. 2026
- [JC2] [Under Review] **Himanshu Gupta, Paul Motter, Bradley Hayes, and Zachary Sunberg**. “Efficient Multi-Robot Motion Planning with Precomputed Translation-Invariant Edge Bundles”. In: *Robotics: Science and Systems (RSS)*. Under review. 2026
- [JC3] [Accepted, Awaiting Publication] **Mel Krusniak, Ofer Dagan, Himanshu Gupta, Benjamin Kraske, Kyle Hollins Wray, and Zachary Sunberg**. “Decisions.jl: Representing and Transforming Decision Problem Classes in Julia”. In: *Autonomous Agents and Multi-Agent Systems (AAMAS)*. Accepted. 2026
- [JC4] [Accepted, Awaiting Publication] **Zakariya Laouar, Qi Heng Ho, and Zachary N. Sunberg**. “Leveraging the Value of Information in POMDP Planning”. In: *International Conference on Automated Planning and Scheduling (ICAPS)*. Accepted. 2026
- [JC5] **Qi Heng Ho, Tyler Becker, Benjamin Kraske, Zakariya Laouar, Martin S. Feather, Federico Rossi, Morteza Lahijanian, and Zachary N. Sunberg**. “Recursively-Constrained Markov Decision Processes”. In: *The 40th Conference on Uncertainty in Artificial Intelligence*. 2024. URL: 
- [JC6] **Qi Heng Ho, Martin S. Feather, Federico Rossi, Zachary N. Sunberg, and Morteza Lahijanian**. “Sound Heuristic Search Value Iteration for Undiscounted POMDPs with Reachability Objectives”. In: *The 40th Conference on Uncertainty in Artificial Intelligence*. 2024. URL: 
- [JC7] **Matt-Heun Hong, Zachary Sunberg, and Danielle Szafir**. “Cieran: Designing Sequential Colormaps with a Teachable Robot”. In: *Human Factors in Computing Systems (CHI)*. 2024. DOI: 10.1145/3613904.3642903
- [JC8] **Himanshu Gupta, Bradley Hayes, and Zachary Sunberg**. “Intention-Aware Navigation in Crowds with Extended-Space POMDP Planning”. In: *Autonomous Agents and Multi-Agent Systems (AAMAS)*. 2022. DOI: 10.48550/arXiv.2206.10028. URL: 
- [JC9] **John Mern, Anil Yildiz, Zachary Sunberg, Tapan Mukerji, and Mykel J. Kochenderfer**. “Bayesian Optimized Monte Carlo Planning”. In: *AAAI Conference on Artificial Intelligence (AAAI)*. 2021. DOI: 10.1609/aaai.v35i13.17411
- [JC10] **Michael H. Lim, Claire J. Tomlin, and Zachary N. Sunberg**. “Sparse Tree Search Optimality Guarantees in POMDPs with Continuous Observation Spaces”. In: *International Joint Conference on Artificial Intelligence (IJCAI)*. 2020. DOI: 10.24963/ijcai.2020/572
- [JC11] **Lasse Peters, David Fridovich-Keil, Claire Tomlin, and Zachary Sunberg**. “Inference-Based Strategy Alignment for General-Sum Differential Games”. In: *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*. 2020. DOI: 10.48550/arXiv.2002.04354. URL: 
- [JC12] **Zachary N. Sunberg and Mykel J. Kochenderfer**. “Online Algorithms for POMDPs with Continuous State, Action, and Observation Spaces”. In: *International Conference on Automated Planning and Scheduling (ICAPS)*. Delft, 2018. DOI: 10.1609/icaps.v28i1.13882. URL: 

Peer Reviewed Conference Publications

Peer reviewed conference publications are selected based on peer review of the entire article, but are considered less rigorous and selective than journal-equivalent conference publications.

- [C1] [Under Review] **Himanshu Gupta, Kelvin Aladum, Bradley Hayes, Nisar Ahmed, and Zachary Sunberg**. “Beyond Fixed Goal Delivery: Online POMDP Planning for Target Interception in Crowds”. In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. Under review. 2026

- [C2] Jiho Lee, Nisar R. Ahmed, Kyle H. Wray, and **Zachary N. Sunberg**. “Rao-Blackwellized POMDP Planning”. In: *IEEE International Conference on Robotics and Automation (ICRA)*. 2025. URL: [🔗](#)
- [C3] Michael Moncton, **Himanshu Gupta**, **Zachary Sunberg**, and Eric Frew. “Autonomous Planning for Targeted Observation of Severe Weather”. In: *Dynamic Data-Driven Application Systems (DDDAS)*. 2024
- [C4] **Zakariya Laouar**, **Qi Heng Ho**, Rayan Mazouz, **Tyler Becker**, and **Zachary N. Sunberg**. “Feasibility-Guided Safety-Aware Model Predictive Control for Jump Markov Linear Systems”. In: *IEEE International Conference on Intelligent Robots and Systems (IROS)*. 2024. doi: 10.1109/IROS58592.2024.10801951
- [C5] Hunter M Ray*, **Zakariya Laouar***, **Zachary N. Sunberg**, and Nisar Ahmed. “Human-Centered Autonomy for UAS Target Search”. In: *IEEE International Conference on Robotics and Automation (ICRA)*. (* equal contribution). 2024. doi: 10.1109/ICRA57147.2024.10611708
- [C6] Sampada Deglurkar, **Michael H. Lim**, **Johnathan Tucker**, **Zachary N. Sunberg**, Aleksandra Faust, and Claire J. Tomlin. “Compositional Learning-based Planning for Vision POMDPs”. In: *Learning for Dynamics & Control (L4DC)*. 2023. doi: 10.48550/arXiv.2112.09456. URL: [🔗](#)
- [C7] **Qi Heng Ho**, **Zachary N. Sunberg**, and Morteza Lahijanian. “Planning with SiMBA: Motion Planning under Uncertainty for Temporal Goals using Simplified Belief Guides”. In: *IEEE International Conference on Robotics and Automation (ICRA)*. London, England, UK, May 2023. doi: 10.1109/ICRA48891.2023.10160897. URL: [🔗](#)
- [C8] **Benjamin D. Kraske**, Anshu Saksena, Anna L. Buczak, and **Zachary Sunberg**. “Explanation through Reward Model Reconciliation using POMDP Tree Search”. In: *IEEE International Conference on Assured Autonomy (ICAA)*. 2023. doi: 10.1109/ICAA58325.2023.00027. URL: [🔗](#)
- [C9] **Qi Heng Ho**, Roland B. Ilyes, **Zachary Sunberg**, and Morteza Lahijanian. “Automaton-Guided Control Synthesis for Signal Temporal Logic Specifications”. In: *IEEE Conference on Decision and Control (CDC)*. 2022. doi: 10.1109/CDC51059.2022.9993090. URL: [🔗](#)
- [C10] **Qi Heng Ho**, **Zachary N. Sunberg**, and Morteza Lahijanian. “Gaussian Belief Trees for Chance Constrained Asymptotically Optimal Motion Planning”. In: *IEEE International Conference on Robotics and Automation (ICRA)*. 2022, pp. 11029–11035. doi: 10.1109/ICRA46639.2022.9812343
- [C11] **Zachary Nolan Sunberg**, Kathryn Anne Wingate, and Lara Buri. “Fair Senior Capstone Project Teaming based on Skills, Preferences, and Friend Groups”. In: *ASEE Annual Conference*. 2021. doi: 10.18260/1-2--37187. URL: [🔗](#)
- [C12] **Michael H. Lim**, Claire J. Tomlin, and **Zachary N. Sunberg**. “Voronoi Progressive Widening: Efficient Online Solvers for Continuous Space MDPs and POMDPs with Provably Optimal Components”. In: *Conference on Decision and Control (CDC)*. 2021. doi: 10.1109/CDC45484.2021.9683490. URL: [🔗](#)
- [C13] Shakeeb G. Ahmad, **Zachary Sunberg**, and Sean Humbert. “APF-PF: Probabilistic Depth Perception for 3D Reactive Obstacle Avoidance”. In: *American Control Conference (ACC)*. 2021. doi: 10.23919/ACC50511.2021.9482894
- [C14] Ekhlas Sonu, **Zachary Sunberg**, and Mykel J. Kochenderfer. “Exploiting Hierarchy for Scalable Decision Making in Autonomous Driving”. In: *Intelligent Vehicles Symposium*. Changshu, 2018. doi: 10.1109/IVS.2018.8500681
- [C15] **Zachary N. Sunberg**, Christopher J. Ho, and Mykel J. Kochenderfer. “The Value of Inferring the Internal State of Traffic Participants for Autonomous Freeway Driving”. In: *American Control Conference (ACC)*. Seattle, 2017. doi: 10.23919/ACC.2017.7963408. URL: [🔗](#)
- [C16] Patrick Slade, Preston Culbertson, **Zachary Sunberg**, and Mykel J. Kochenderfer. “Simultaneous Active Parameter Estimation and Control using Sampling-based Bayesian Reinforcement Learning”. In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2017. doi: 10.1109/IROS.2017.8202242. URL: [🔗](#)
- [C17] **Zachary Sunberg**, Mykel J. Kochenderfer, and Marco Pavone. “Optimized and Trusted Collision Avoidance for Unmanned Aerial Vehicles using Approximate Dynamic Programming”. In: *IEEE*

International Conference on Robotics and Automation (ICRA). Stockholm, 2016. DOI: 10.1109/ICRA.2016.7487280. URL: 

[C18] **Zachary Sunberg**, Suman Chakravorty, and Richard Erwin. "Information space sensor tasking for Space Situational Awareness". In: *American Control Conference (ACC)*. June 2014, pp. 79–84. DOI: 10.1109/ACC.2014.6858922

[C19] **Zachary Sunberg**, Nathaniel Miller, and Jonathan Rogers. "A Real Time Expert Control System for Helicopter Autorotation". In: *70th Forum of the American Helicopter Society*. Montreal, 2014

Conference Publications with Abstract-only Review.....

[A1] **Tyler Becker** and **Zachary N. Sunberg**. "Strategic Satellite Custody Maintenance with AlphaZero". In: *Advanced Maui Optical and Space Surveillance Technologies Conference (AMOS)*. 2025. URL: 

[A2] **Ofer Dagan**, **Tyler Becker**, and **Zachary N. Sunberg**. "Hypothesis-Driven Sensor Tasking for Space Domain Awareness". In: *Advanced Maui Optical and Space Surveillance Technologies Conference (AMOS)*. 2025. URL: 

[A3] **Johnathan Tucker**, **Jackson Wagner**, and **Zachary Sunberg**. "Adaptive Stress Testing Applied To Space Domain Awareness Systems". In: *Advanced Maui Optical and Space Surveillance Technologies (AMOS)*. 2022. URL: 

[A4] **Tyler Becker** and **Zachary Sunberg**. "Imperfect Information Games and Counterfactual Regret Minimization in Space Domain Awareness". In: *Advanced Maui Optical and Space Surveillance Technologies (AMOS)*. 2022. URL: 

[A5] Hunter M. Ray, Nicholas Conlon, **Zachary Sunberg**, and Nisar R. Ahmed. "User Preference Elicitation for Unmanned Aircraft System Collaborative Search". In: *AIAA SCITECH Forum*. 2022, p. 2343. DOI: 10.2514/6.2022-2343

[A6] **Zachary Sunberg** and Jonathan Rogers. "A Fuzzy Logic-Based Controller for Helicopter Autorotation". In: *AIAA Aerospace Sciences Meeting*. Grapevine, 2013. DOI: 10.2514/6.2013-1150

Extended Abstracts.....

[E1] **Tyler Becker** and **Zachary Sunberg**. "Bridging the Gap between Partially Observable Stochastic Games and Sparse POMDP Methods". In: *Autonomous Agents and Multi-Agent Systems AAMAS*. Accepted as extended abstract. 2025

[E2] **Ofer Dagan**, **Tyler Becker**, and **Zachary N. Sunberg**. "Resolving Multiple-Dynamic Model Uncertainty in Hypothesis-Driven Belief-MDPs". In: *Autonomous Agents and Multi-Agent Systems (AAMAS)*. Extended abstract. 2025

Open Source Software

POMDPs.jl [2015 – present]

<https://github.com/JuliaPOMDP/POMDPs.jl>

Interface for defining continuous and discrete, fully and partially observable Markov decision processes along with a suite of state-of-the art solvers written in Julia and C++ [J12]. 33 direct package contributors, 44 dependent packages, 650 GitHub stars, 430 monthly downloads according to juliahub.com.

satisfaculty [2025 – present]

<https://www.satisfaculty.com>

Python package for optimizing teaching schedules using integer linear programming. Used to create Smead AES Fall 2026 schedule.

ProjectAssigner.jl [2020]

<https://github.com/zsunberg/ProjectAssigner.jl>

Software for optimally assigning student teams to projects based on preferences, friend groups, and skills [C11]. Used to assign senior project groups in the AES department from 2020 to 2024.

Research Funding

PI: CAREER: Game-theoretic Online Planning in Partially Observable Domains [2024 – 2029]
National Science Foundation
\$599,985

CU PI: DECODE AI: Deception and Counter-Deception in Artificial Intelligence [2024 – 2028]
Office of Naval Research
\$4,150,000 (total award) | \$1,014,000 (my share); PI: Matthew Hale, Georgia Tech, Co-Is: Ufuk Topku, UT Austin; Ruimeng Hu, UCSB

CU PI: Collaborative Research: Alternative Leaf Water use Strategies in Hot Environments [2023 – 2024]
National Science Foundation
\$944,823 (total award) | \$99,976 (my share), PI: Ben Blonder, UC Berkeley; (I did not participate in securing this grant, but was subcontracted to work on it later.)

PI: Human Centered Autonomy for Dynamic sUAS Target Search Operations [2023 – 2026]
NSF Center for Autonomous Air Mobility and Sensing (CAAMS) Sub-award
\$287,000 (total award) | \$152,000 (my share, approx.); Co-I: Nisar Ahmed, CU Boulder

Co-I: Multi-Phenomenological, Autonomous ... Decision Support [2023 – 2028]
Air Force Office of Scientific Research
\$5,000,000 (total award, approx.) | \$500,000 (my share, approx.), PI: Marcus Holtzinger, CU Boulder; Co-Is: K. Terry Alfriend (TAMU), Scott Palo, Kyle DeMars (TAMU), John Junkins (TAMU), Karen Feigh (Georgia Tech) | Full title: Multi-Phenomenological, Autonomous, and Understandable SDA and XDA Decision Support

Co-I: IUCRC Phase I: Center for Autonomous Air Mobility and Sensing (CAAMS) [2022 – 2027]
National Science Foundation
\$2,210,225 (total award) | \$0 (my direct share - all funds shared for travel and administrative costs), PI: Eric Frew, CU Boulder; Co-Is: Nisar Ahmed, Morteza Lahijanian, and Sriram Sankaranarayanan, CU Boulder; This grant establishes the IUCRC; sub-awards funded through industry contributions are listed separately.

Co-I: Dispersed Autonomy for Marsupial Aerial Robot Teams [2022 – 2025]
National Science Foundation | National Robotics Institute (Collaborative Research)
\$1,045,429 (total award) | \$300,000 (my share, approx.), PI: Eric Frew; Co-Is: Brian Argrow, Adam Houston (University of Nebraska, Lincoln)

PI: HIPPO (Human-Informed Planning with Probabilistic Observations) [2022-2023]
NSF Center for Autonomous Air Mobility and Sensing (CAAMS) Sub-award
\$68,000

CU PI: SURP: Fast planning under uncertainty with operational and safety guarantees [2022-2024]
NASA Jet Propulsion Laboratory
\$180,000 (total award) | \$150,000 (my share), PI: Federico Rossi, JPL; Co-I: Morteza Lahijanian, CU Boulder

CU PI: Elektra: Naval Defensive Resource Allocation through POMDP Optimization [2021 – 2023]
Office of Naval Research (Subcontract of Johns Hopkins University Applied Physics Lab)
\$169,805 | Subcontract on larger award of unknown total amount

Co-I: L3Harris Modern Analytics for Mission Applications (LMA2) [2021-2023]
L3-Harris Technologies
\$1,495,048 (total award) | \$350,000 (my share, approx.), PI: Marcus Holzinger, CU Boulder (I did not participate in securing this grant, but began work on it later.)

CU PI: Full Stack Planning and Control under Uncertainty [2021 – 2022]
NSF Center for Unmanned Aircraft Systems (C-UAS) Sub-award
\$130,000 (total award) | \$65,000 (my share), Co-PI Ella Atkins, Univ. of Michigan

PI: POMDP Algorithms for In-flight Learning in Emergencies [2020 – 2021]
NSF Center for Unmanned Aircraft Systems (C-UAS) Sub-award
\$60,000

Invited Talks and Panels

- [I1] **Technion – Israel Institute of Technology (Online):** “Breaking the curse of dimensionality in POMDPs and games with sampling-based online planning”. 2025
- [I2] **Autonomy Talks Series, Massachusetts Institute of Technology (MIT) (Online):** “Breaking the curse of dimensionality in POMDPs with sampling-based online planning”. 2024. <https://www.youtube.com/watch?v=XDIyX0tanjk>
- [I3] **University of California, San Diego:** “Breaking the curse of dimensionality in POMDPs with sampling-based online planning”. 2024
- [I4] **University of Illinois Urbana-Champaign (UIUC):** “Safe and efficient autonomy in the face of uncertainty”. 2022
- [I5] **University of Texas at Austin:** “Safe and efficient autonomy in the face of uncertainty”. 2022
- [I6] **Korea Advanced Institute of Science and Technology (KAIST) (Online):** “Safe and efficient autonomy in the face of state and interaction uncertainty”. 2022
- [I7] **AIAA Rocky Mountain Annual Technical Symposium:** “Machine Learning in Aerospace Systems”. 2021 (panel)
- [I8] **NASA Jet Propulsion Lab, Pasadena, CA:** “Scalable online POMDP planning for safe and efficient autonomy”. 2021
- [I9] **Johns Hopkins University Applied Physics Lab, Laurel, MD:** “Some Recent Advances in Online POMDP Algorithms”. 2021
- [I10] **SRI International, Palo Alto, CA:** “Safety and Efficiency for Autonomous Vehicles through Online Learning”. 2019
- [I11] **Washington State University, Pullman, WA:** “Safety and Efficiency for Autonomous Vehicles through Online Learning”. 2019
- [I12] **University of Colorado, Boulder, CO:** “Safety and Efficiency for Autonomous Vehicles through Online Learning”. 2019
- [I13] **Renault-Nissan Research, Sunnyvale, CA:** “Safety and Efficiency in Autonomous Vehicles through POMDP Planning”. 2018
- [I14] **Lyft Level 5, Palo Alto, CA:** “Safety and Efficiency in Autonomous Vehicles through Planning with Uncertainty”. 2018
- [I15] **Makani, Alameda, CA:** “Algorithms for Uncertain, Non-convex Control Problems in the Real World”. 2018
- [I16] **Indeed, San Francisco, CA:** “Safety and Efficiency in Autonomous Vehicles through Planning with Uncertainty”. 2018

Other Presentations

JuliaCon 2021: “POMDPs.jl and Interactive Assignments in Julia”. 2021
Julia in Controls Workshop, ACC, Seattle, WA: “POMDPs.jl”. 2017
Open Source Software for Decision Making (OSS4DM), Stanford, CA: “POMDPs.jl - Challenges and Lessons Learned”. 2017

Teaching

At CU Boulder

ASEN 3728/3128 Aircraft Dynamics [Fa 2022, Sp 2024, Sp 2025]
University of Colorado Boulder, typical enrollment: 140
Required Junior-level course in aircraft dynamics, stability, and control.
Team taught with Prof. Eric Frew in 2022.
Open-source course materials: <https://github.com/zsunberg/Aircraft-Dynamics-Materials>

ASEN 5264 Decision Making under Uncertainty [Sp 2020–Sp 2026]

University of Colorado Boulder, typical enrollment: 30-60

New entry-level graduate course about decision making under uncertainty created by me.

Open-source course materials: <https://github.com/zsunberg/CU-DMU-Materials>

Open-source companion software package: <https://github.com/zsunberg/DMUStudent.jl>

ASEN 6519: Optimization: Applications and Algorithms [Fa 2025]

University of Colorado Boulder, typical enrollment: 10-15

New advanced graduate course that surveys optimization techniques created by me.

Open-source course materials: <https://github.com/zsunberg/CU-Optimization-Materials>

ASEN 6519 Advanced Survey of Sequential Decision Making [Fa 2021, Fa 2023, Fa 2025]

University of Colorado Boulder, typical enrollment: 10-15

New advanced graduate course that surveys recent advances in decision making under uncertainty created by me.

ASEN 4018/4028 Senior Design Project [Fa 2020–Sp 2021]

University of Colorado Boulder, typical enrollment: 250-300, team size: 9-12

Capstone senior design course. I mentored two teams and created new optimization-based approach (see Open Source Software) for fairly creating teams based on student preferences [C11].

Prior to CU Boulder.....

Army High Performance Computing Summer Institute [June 2017]

Stanford University, Stanford, CA

Developed and taught a 5 lecture course about decision making under uncertainty for college students.

Stanford Artificial Intelligence Lab OutReach Summer (SAILORS, now AI4ALL) [2015-2017]

Stanford University, Stanford, CA

Developed and taught a 2 week course and project for high school students that included programming robots for optical line following and using Dijkstra's algorithm to find the shortest path on a road network; only project mentor to serve all three years of the program. <http://ai-4-all.org/>

AA-228/CS-238 Decision Making Under Uncertainty [Autumn 2016]

Stanford University, Stanford, CA

Head course assistant for a class of around 200; developed problems for midterm project; gave guest lectures on the POMDPs.jl framework and autonomous driving research; project software reused at Iowa State University.

Advising and Mentoring

Postdoctoral Scholars.....

Ofer Dagan [2024 – 2025]

Founding Machine Learning Engineer, Eiro Inc.

Graduated Ph.D. Thesis Advisees.....

Qi Heng Ho (Co-advised with Morteza Lahijanian) [2020 – 2025]

Assistant Professor, Virginia Tech

Hyun Jae (Michael) Lim (Co-advised with Claire Tomlin at U.C. Berkeley) [2020 – 2023]

AI Scientist, C3.ai

Current Ph.D. Thesis Advisees.....

Tyler Becker (Comprehensive exam spring 2026) [Fall 2020 – Present]

Ben Kraske (Comprehensive exams spring 2026, Received NSF GRFP) [Fall 2020 – Present]

Zakariya Laouar (Prelim exam fall 2022) [Fall 2021 – Present]

Himanshu Gupta (Prelim exam fall 2023) [Fall 2020 (Started as MS) – Present]

Lisa Ventura (Prelim exam fall 2025) [Fall 2024 – Present]

Mark Boyer (Prelim exam fall 2025) [Fall 2024 – Present]

Mel Krusniak [Spring 2026 – Present]

Graduated M.S. Thesis Advisees.....

William Pope [Spring 2022 – Fall 2022]
U.S. Space Force Officer

Independent Study Advisees.....

Austin Monell [2023 – 2024]
Johnathan Tucker (Received NSF GRFP under my direction) [Fall 2020 – Summer 2022]
Saurabh Mishra [Spring 2020]

Ph.D. Comprehensive Exam and Defense Committees.....

Sangwoo Moon	Aastha Acharya
Charles (Luke) Burks	Shohei Wakayama
Neha Garg (Nat. Univ. of Singapore, External Examiner)	Marcus Lapeyrolerie (Univ. of California, Berkeley, Qualls)
Ramya Kanlapuli	John R. Martin
Katherine Glasheen	Adam Herrmann
Andrew Mills	Camron (Alex) Hirst
Chandrakanth Venigalla	Hunter Ray
Sam Fedeler	Michael Klonowski
John Jackson	Mark Stephenson
Prashin Sharma (Univ. of Michigan)	Michael Moncton
Shakeeb Ahmad	Matt Budd (University of Oxford)
John Mern (Stanford Univ.)	

M.S. Thesis Committees.....

Lasse Peters (TU Hamburg)	Jamison Mcginley
Wyatt Raich	Rio McMahon
Cody Charland	Eli Kravitz
Akash Ratheesh	Abdoulaye Diallo

Service

Co-Organizer of <i>Faculty Sanity</i> University-Wide Happy Hour	[2024–present]
Undergraduate Curriculum Committee	[2024–present]
Onboarding mentor for new faculty member	[2023–present]
Undergraduate Operations Committee	[2023–present]
Optimized undergraduate teaching assignments for course scheduling for the department.	
Autonomous Systems Faculty Search Committee	[2023–2025]
Graduate Program Committee	[2020–2023]
<i>Autonomous Systems Lead</i>	
Served on Preliminary Exam Subcommittee, Revised MS Admissions Criteria	

Outreach and Inclusion

Speaker for Tuskegee Airmen outreach event at CU	[2021, 2022, 2025]
Smead aerospace career panelist	[2023]
Rising Stars in Aerospace organizing committee member	[2022]
Invited Speaker for AIAA Movie Night & Technical Discussion: 2001: A Space Odyssey	[2021]

Conference and Workshop Organization

Inference and Decision Making for Autonomous Vehicles (IDMAV)	[2023]
<i>Workshop at Robotics: Science and Systems (RSS)</i>	
Co-organizer with Christoffer Heckman, Han-Lim Choi (KAIST), and students	

Strategic multi-agent interactions: game theory for robot learning and decision making [2022]
Workshop at the Conference on Robotic Learning (CoRL)
Co-organizer with David Fridovich-Keil (Univ. of Texas), Negar Mehr (Univ. of Illinois), and Forrest Laine (Vanderbilt)

Academic Peer Review and Editing

Area chair for Robotics: Science and Systems (RSS) Conference [2025 – 2026]
Guest editor for the AIAA Journal of Aerospace Information Systems [2020 – 2022]

Reviewer: I have reviewed manuscripts for the following journals and conferences:

- | | |
|---|---|
| AAAI Conference on Artificial Intelligence | IEEE Robotics and Automation Letters |
| ACM Transactions on Mathematical Software | IEEE Transactions on Cybernetics |
| AIAA Journal of Guidance, Control, and Dynamics | IEEE Transactions on Intelligent Transportation Systems |
| AIAA Journal of Aerospace Information Systems | IEEE Transactions on Intelligent Vehicles |
| American Control Conference (ACC) | Intelligent Transportation Systems Conference (ITSC) |
| American Geophysical Union (AGU) Books | Intl. Journal of Robotics Research (IJRR) |
| Journal of the American Helicopter Society | Intl. Symposium on Robotics Research |
| Autonomous Robots | Intl. Joint Conference on Artificial Intelligence (IJCAI) |
| Artificial Intelligence | Learning for Decision and Control Conference (L4DC) |
| Journal of Artificial Intelligence Research | Operations Research |
| Field Robotics | R (programming language) Journal |
| IEEE Conference on Decision and Control (CDC) | Robotics, Science and Systems (RSS) |
| IEEE Intl. Conf. on Robotics and Automation (ICRA) | Conf. on Uncertainty in Artificial Intelligence (UAI) |
| IEEE Intl. Conf. on Intelligent Robots and Systems (IROS) | |

Proposal Review

Proposal Reviewer for Israel Science Foundation (ISF) [2026]
NSF Computer and Information Science and Engineering (CISE) Proposal Review Panelist [2023]
Army Research Office (ARO) Proposal Reviewer [2023]